Product Technical Manual
Component: Web Console
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Troubleshooting:
- Symptom: service fails to start
- Error code: E610

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- invalid config value
- expired certificate

Recovery Actions:
- rotate certificate and restart
- rotate certificate and restart
- rotate certificate and restart

Fingerprint: 32383fe6597f2de6bec3a2ffe7815adf

Upgrade Notes:
- From version: 1.3.5
- To version: 1.4.1
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] confirm maintenance window
[] prepare rollback artifact
[] validate free disk space
[] notify users
[] export config snapshot
[] disable non-essential jobs
[] take a full backup
[] verify compatibility matrix
[] verify certificates are valid
[] run pre-upgrade diagnostics

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl status
- Verification: no errors in last 5 minutes

Fingerprint: c3d93297a8b4365c25116a242ce6656b

Task Overview:
- Task: Upgrade
- Applies to: Connector
- Prerequisites: database reachable

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: cache.size_mb = tls
3. Restart service: product-api
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 166 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restart
- productctl diagnostics

Fingerprint: 510cf3a8c0c6bd7e87fe5d563b660a80

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E191

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- insufficient permissions
- insufficient permissions

Recovery Actions:
- clear cache directory
- reset config to baseline
- increase memory limit and redeploy

Fingerprint: ea1d8d21f36d8dd21eda63cb537ca505

Upgrade Notes:
- From version: 1.3.5
- To version: 1.4.1
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] export config snapshot
[] verify compatibility matrix
[] confirm maintenance window
[] disable non-essential jobs
[] validate free disk space
[] stop services cleanly
[] record current version

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl status
- Verification: metrics endpoint responds

Fingerprint: f6be7b4730a3764c1df540b66b30bb43

Task Overview:
- Task: Backup
- Applies to: Web Console
- Prerequisites: admin access

Procedure:
1. Validate environment: ports open
2. Set configuration: db.pool_max = 8080
3. Restart service: product-ui
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 53 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl status
- productctl diagnostics

Fingerprint: c99cffc049aa46a3a741512ffe117b16

Troubleshooting:
- Symptom: authentication loop
- Error code: E433

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- expired certificate
- invalid config value

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- rotate certificate and restart

Fingerprint: b36e028e1180c3666c994c35dd428bb9

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] prepare rollback artifact
[] confirm maintenance window
[] record current version
[] take a full backup
[] disable non-essential jobs
[] verify certificates are valid
[] export config snapshot
[] stop services cleanly
[] notify users
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restart
- Verification: metrics endpoint responds

Fingerprint: d81afd2d6ed06446452996f010f7cfc6

Task Overview:
- Task: Collect Diagnostics
- Applies to: Control Plane
- Prerequisites: database reachable

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: cache.size_mb = alpha,beta
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 76 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl diagnostics

Fingerprint: d38bc05deece6cda1fb8f12ba3cb725c

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E157

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- incompatible dependency version
- expired certificate

Recovery Actions:
- re-run installer with admin rights
- clear cache directory
- reset config to baseline

Fingerprint: 4bf6d83f6abc40ba5c6a4f4879bed3a9

Upgrade Notes:
- From version: 1.4.1
- To version: 1.2.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] verify certificates are valid
[] stop services cleanly
[] record current version
[] disable non-essential jobs
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] validate free disk space
[] notify users

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl diagnostics
- Verification: UI loads and login works

Fingerprint: 786b5745700995ec31d0382b9a311ba5

Task Overview:
- Task: Install
- Applies to: Control Plane
- Prerequisites: database reachable

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: db.pool_max = 40
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 79 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl backup --dest backups

Fingerprint: 100fd7c058bb47fb24041774de668eb7

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E332

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- expired certificate
- network policy block

Recovery Actions:
- re-run installer with admin rights
- clear cache directory
- increase memory limit and redeploy

Fingerprint: 40502df87b6c35ec18495bf80f2fea87

Upgrade Notes:
- From version: 1.3.5
- To version: 1.2.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] notify users
[] record current version
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] verify certificates are valid
[] disable non-essential jobs
[] verify compatibility matrix
[] export config snapshot
[] stop services cleanly
[] take a full backup

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl status
- Verification: no errors in last 5 minutes

Fingerprint: 87ab357c9e070942dd79e05782b3a189

Task Overview:
- Task: Tune Performance
- Applies to: Database Layer
- Prerequisites: admin access

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: log.level = 512
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 46 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl status
- productctl restore --src backups

Fingerprint: 92a227a90cd55d73b283cea30583a529

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E456

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- network policy block
- network policy block

Recovery Actions:
- increase memory limit and redeploy
- clear cache directory
- re-run installer with admin rights

Fingerprint: 6eb5f53efc172a8de0b7e8ba9137b942

Upgrade Notes:
- From version: 2.1.2
- To version: 1.3.5
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] take a full backup
[] prepare rollback artifact
[] confirm maintenance window
[] disable non-essential jobs
[] record current version
[] stop services cleanly
[] export config snapshot
[] validate free disk space
[] verify certificates are valid
[] run pre-upgrade diagnostics

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl diagnostics
- Verification: no errors in last 5 minutes

Fingerprint: 568165cf8aecf86dd75d0f5569a31cf6

Task Overview:
- Task: Upgrade
- Applies to: Telemetry Module
- Prerequisites: supported OS image

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: cache.size_mb = 40
3. Restart service: product-api
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 67 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl status
- productctl restore --src backups

Fingerprint: e9f8c971e305cb5827bfc86f8cfb0193

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E915

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- rotate certificate and restart
- rotate certificate and restart
- clear cache directory

Fingerprint: d0de27f202ed6951fe9ad5ad921db1d1

Upgrade Notes:
- From version: 2.0.0
- To version: 2.1.2
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] validate free disk space
[] verify certificates are valid
[] stop services cleanly
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] take a full backup
[] disable non-essential jobs
[] record current version
[] notify users
[] confirm maintenance window

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl status
- Verification: status endpoint returns OK

Fingerprint: 1b916d9eb581ca52297089398a53bdcb

Task Overview:
- Task: Upgrade
- Applies to: Web Console
- Prerequisites: database reachable

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: server.port = info
3. Restart service: product-agent
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 130 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl status

Fingerprint: 68b57a6352b3f4116196cbbb3620aab6

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E670

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- incompatible dependency version
- insufficient permissions

Recovery Actions:
- clear cache directory
- increase memory limit and redeploy
- increase memory limit and redeploy

Fingerprint: cefdc8be01e562f2010c2368ab2694da

Upgrade Notes:
- From version: 1.2.0
- To version: 1.4.1
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] verify certificates are valid
[] prepare rollback artifact
[] record current version
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] notify users
[] validate free disk space
[] stop services cleanly

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restore --src backups
- Verification: metrics endpoint responds

Fingerprint: 75173428eb025b3b1e2d9d9289381938

Task Overview:
- Task: Configure
- Applies to: CLI
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: db.pool_max = 40
3. Restart service: product-agent
4. Verify health: UI loads and login works

Notes:
- Default timeout: 49 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl status

Fingerprint: 76335c692ba4a62a881af15464c90da6

Troubleshooting:
- Symptom: service fails to start
- Error code: E987

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- invalid config value
- incompatible dependency version

Recovery Actions:
- reset config to baseline
- rotate certificate and restart
- reset config to baseline

Fingerprint: 166ff5ddbdaf3372f05e000e09f24821

Upgrade Notes:
- From version: 2.0.0
- To version: 1.3.5
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] validate free disk space
[] confirm maintenance window
[] notify users
[] run pre-upgrade diagnostics
[] export config snapshot
[] record current version
[] verify compatibility matrix
[] take a full backup
[] stop services cleanly
[] verify certificates are valid

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl status
- Verification: status endpoint returns OK

Fingerprint: cb3e195a082df23871cc199f486613e0

Task Overview:
- Task: Backup
- Applies to: Telemetry Module
- Prerequisites: DNS configured

Procedure:
1. Validate environment: ports open
2. Set configuration: feature.flags = 8080
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 155 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl restore --src backups

Fingerprint: f34b822a5ae820c908d2f7571423aa67

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E713

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- re-run installer with admin rights
- clear cache directory
- re-run installer with admin rights

Fingerprint: 5644821c5bc486ddd1274d8f4691bf9f

Upgrade Notes:
- From version: 1.3.5
- To version: 1.2.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] verify certificates are valid
[] notify users
[] confirm maintenance window
[] prepare rollback artifact
[] verify compatibility matrix
[] record current version
[] disable non-essential jobs
[] take a full backup
[] validate free disk space
[] stop services cleanly

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl status
- Verification: UI loads and login works

Fingerprint: 7b03a861fd2f54cb3c71dc082ad9791e

Task Overview:
- Task: Backup
- Applies to: Database Layer
- Prerequisites: DNS configured

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: db.pool_max = info
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 129 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl status
- productctl restart

Fingerprint: b7db73ad355ef64cdca93616586f4f2c

Troubleshooting:
- Symptom: high CPU usage
- Error code: E408

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- expired certificate
- network policy block

Recovery Actions:
- rotate certificate and restart
- reset config to baseline
- reset config to baseline

Fingerprint: b7ab2374d60d71fd91c29ed39c87e69a

Upgrade Notes:
- From version: 1.4.1
- To version: 1.4.1
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] confirm maintenance window
[] verify certificates are valid
[] notify users
[] record current version
[] stop services cleanly
[] disable non-essential jobs
[] validate free disk space
[] take a full backup
[] run pre-upgrade diagnostics
[] export config snapshot

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: ad72f2e53451c8d6648e9bf945ea783d

Task Overview:
- Task: Backup
- Applies to: Scheduler
- Prerequisites: DNS configured

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: server.port = alpha,beta
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 143 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl diagnostics

Fingerprint: dfbf50f580bb85a489c40471d308bab0

Troubleshooting:
- Symptom: authentication loop
- Error code: E211

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- network policy block
- expired certificate

Recovery Actions:
- reset config to baseline
- re-run installer with admin rights
- rotate certificate and restart

Fingerprint: d1c2458a1ffe8276adca2418af9e6203

Upgrade Notes:
- From version: 1.4.1
- To version: 2.1.2
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] validate free disk space
[] prepare rollback artifact
[] disable non-essential jobs
[] take a full backup
[] stop services cleanly
[] notify users
[] export config snapshot
[] confirm maintenance window

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl restart
- Verification: status endpoint returns OK

Fingerprint: 2301c3c55cbbce4d65c97584ccdc08cc

Task Overview:
- Task: Upgrade
- Applies to: Agent
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: telemetry.enabled = alpha,beta
3. Restart service: telemetry
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 86 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl restart
- productctl status

Fingerprint: 24733942e05e3e29bfc272f1b53bcb20

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E501

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- invalid config value
- network policy block

Recovery Actions:
- rotate certificate and restart
- re-run installer with admin rights
- clear cache directory

Fingerprint: 8ddb612ced33c4e4cf2599bcd55a1093

Upgrade Notes:
- From version: 1.4.1
- To version: 2.1.2
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] prepare rollback artifact
[] notify users
[] verify certificates are valid
[] verify compatibility matrix
[] stop services cleanly
[] validate free disk space
[] take a full backup
[] confirm maintenance window

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restore --src backups
- Verification: status endpoint returns OK

Fingerprint: d7ea8d5e04d6e70cfba07f76e233fd28

Task Overview:
- Task: Upgrade
- Applies to: Database Layer
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: auth.mode = 512
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 106 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl diagnostics
- productctl restart

Fingerprint: 6d8cf0f6186a435b58ea1487fb80a462

Troubleshooting:
- Symptom: service fails to start
- Error code: E619

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- incompatible dependency version
- insufficient permissions

Recovery Actions:
- rotate certificate and restart
- reset config to baseline
- rotate certificate and restart

Fingerprint: 29f2f449e565cfbfa8b26f4a391e9f27

Upgrade Notes:
- From version: 1.2.0
- To version: 2.1.2
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] take a full backup
[] validate free disk space
[] verify compatibility matrix
[] notify users
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] export config snapshot
[] confirm maintenance window
[] disable non-essential jobs
[] stop services cleanly

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: 04be85e3e9d247971ab2dc4ece49f8c6

Task Overview:
- Task: Upgrade
- Applies to: Control Plane
- Prerequisites: admin access

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: cache.size_mb = true
3. Restart service: product-ui
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 68 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl restart
- productctl restore --src backups

Fingerprint: 830ced66c14623d59900d9fc379e8e11

Troubleshooting:
- Symptom: authentication loop
- Error code: E743

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- network policy block
- incompatible dependency version

Recovery Actions:
- rotate certificate and restart
- clear cache directory
- reset config to baseline

Fingerprint: 6cce51f3015582b5835c5e5c5cd51a8f

Upgrade Notes:
- From version: 2.1.2
- To version: 1.3.5
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] verify certificates are valid
[] confirm maintenance window
[] export config snapshot
[] prepare rollback artifact
[] validate free disk space
[] stop services cleanly
[] take a full backup
[] notify users
[] record current version

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl status
- Verification: no errors in last 5 minutes

Fingerprint: 781b9caafba8c94a3ba219e6835a6bdb

Task Overview:
- Task: Rotate Credentials
- Applies to: Control Plane
- Prerequisites: database reachable

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: db.pool_max = 8080
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 35 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl diagnostics

Fingerprint: 42dded75ad79594a7f6fd3b00bcbfbf3

Troubleshooting:
- Symptom: authentication loop
- Error code: E867

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- expired certificate
- network policy block

Recovery Actions:
- re-run installer with admin rights
- clear cache directory
- clear cache directory

Fingerprint: f5194ce0ecfbc46e06fed7d463f13fd2

Upgrade Notes:
- From version: 2.1.2
- To version: 2.1.2
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] stop services cleanly
[] confirm maintenance window
[] notify users
[] export config snapshot
[] verify compatibility matrix
[] validate free disk space
[] record current version
[] take a full backup
[] run pre-upgrade diagnostics
[] prepare rollback artifact

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restore --src backups
- Verification: no errors in last 5 minutes

Fingerprint: bc2bc5a21b43ba7a78f5974599aea208

Task Overview:
- Task: Collect Diagnostics
- Applies to: Scheduler
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: log.level = info
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 265 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl status

Fingerprint: 990bc0968794360efb7820bb4e2ad9d0

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E125

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- invalid config value
- network policy block

Recovery Actions:
- increase memory limit and redeploy
- clear cache directory
- reset config to baseline

Fingerprint: d0918610bb2f8affe95e3f4720061058

Upgrade Notes:
- From version: 1.2.0
- To version: 1.4.1
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] verify certificates are valid
[] take a full backup
[] export config snapshot
[] prepare rollback artifact
[] record current version
[] run pre-upgrade diagnostics
[] confirm maintenance window
[] validate free disk space
[] notify users
[] disable non-essential jobs

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl restore --src backups
- Verification: status endpoint returns OK

Fingerprint: 8e53a4576399fcbb38030d9f1a04e0ce

Task Overview:
- Task: Collect Diagnostics
- Applies to: Scheduler
- Prerequisites: database reachable

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: feature.flags = info
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 92 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl backup --dest backups
- productctl restore --src backups

Fingerprint: 0abcbcefe6ea5e23042e5be2d9733754

Troubleshooting:
- Symptom: service fails to start
- Error code: E149

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- network policy block
- insufficient permissions

Recovery Actions:
- re-run installer with admin rights
- re-run installer with admin rights
- rotate certificate and restart

Fingerprint: 642a556ac52203c1d132a09c2884d202

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] validate free disk space
[] disable non-essential jobs
[] prepare rollback artifact
[] verify compatibility matrix
[] take a full backup
[] verify certificates are valid
[] confirm maintenance window
[] notify users
[] export config snapshot
[] stop services cleanly

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restore --src backups
- Verification: status endpoint returns OK

Fingerprint: c35748f9e428cf4692af8e0dc1af9535

Task Overview:
- Task: Collect Diagnostics
- Applies to: Web Console
- Prerequisites: admin access

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: telemetry.enabled = alpha,beta
3. Restart service: product-api
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 40 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl restore --src backups

Fingerprint: 54fe31dd68ec4bec42cdbc80f7c31dce

Troubleshooting:
- Symptom: authentication loop
- Error code: E120

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- network policy block
- network policy block

Recovery Actions:
- rotate certificate and restart
- re-run installer with admin rights
- increase memory limit and redeploy

Fingerprint: 745fef6eb22549867ced5826740e12f8

Upgrade Notes:
- From version: 1.4.1
- To version: 2.0.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] stop services cleanly
[] validate free disk space
[] record current version
[] disable non-essential jobs
[] export config snapshot
[] verify certificates are valid
[] prepare rollback artifact
[] notify users
[] verify compatibility matrix
[] confirm maintenance window

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: status endpoint returns OK

Fingerprint: cc30d7bb864360ce4b891f713f146b5a

Task Overview:
- Task: Upgrade
- Applies to: CLI
- Prerequisites: TLS certificates ready

Procedure:
1. Validate environment: ports open
2. Set configuration: auth.mode = 512
3. Restart service: product-api
4. Verify health: UI loads and login works

Notes:
- Default timeout: 175 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl backup --dest backups
- productctl restore --src backups

Fingerprint: 50985790f6cf2618a9a78af3362cd0d2

Troubleshooting:
- Symptom: high CPU usage
- Error code: E317

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- insufficient permissions
- invalid config value

Recovery Actions:
- re-run installer with admin rights
- re-run installer with admin rights
- increase memory limit and redeploy

Fingerprint: a1663cda98029b8924075ab9a6963159

Upgrade Notes:
- From version: 1.4.1
- To version: 1.2.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] stop services cleanly
[] prepare rollback artifact
[] verify compatibility matrix
[] export config snapshot
[] take a full backup
[] validate free disk space
[] verify certificates are valid
[] run pre-upgrade diagnostics
[] record current version
[] disable non-essential jobs

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restart
- Verification: UI loads and login works

Fingerprint: 71dca00aa98b4dc106663ef9e9afa541

Task Overview:
- Task: Restore
- Applies to: Control Plane
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: db.pool_max = alpha,beta
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 88 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restart
- productctl backup --dest backups

Fingerprint: 38db8bd1aa80e72f102595d7c885c43b

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E819

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- incompatible dependency version
- invalid config value

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- rotate certificate and restart

Fingerprint: 6af6a61c34cd128dab3d0081d607e77c

Upgrade Notes:
- From version: 2.1.2
- To version: 2.1.2
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] notify users
[] export config snapshot
[] verify certificates are valid
[] verify compatibility matrix
[] disable non-essential jobs
[] stop services cleanly
[] confirm maintenance window
[] validate free disk space
[] record current version
[] run pre-upgrade diagnostics

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: 8fbf71803fc37b5718d56cb2d3cb2740

Task Overview:
- Task: Configure
- Applies to: Agent
- Prerequisites: database reachable

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: cache.size_mb = info
3. Restart service: telemetry
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 39 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl diagnostics
- productctl backup --dest backups

Fingerprint: 96d9645d0f3897d04097e373cb8c2913

Troubleshooting:
- Symptom: authentication loop
- Error code: E881

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- invalid config value
- insufficient permissions

Recovery Actions:
- clear cache directory
- increase memory limit and redeploy
- re-run installer with admin rights

Fingerprint: 584e9d5194e7ba99cb2671892d63e28b

Upgrade Notes:
- From version: 2.0.0
- To version: 1.4.1
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] take a full backup
[] prepare rollback artifact
[] disable non-essential jobs
[] stop services cleanly
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] export config snapshot
[] validate free disk space

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl restore --src backups
- Verification: metrics endpoint responds

Fingerprint: 53035d73022633c8fc00d2d6db25aa68

Task Overview:
- Task: Tune Performance
- Applies to: Connector
- Prerequisites: DNS configured

Procedure:
1. Validate environment: ports open
2. Set configuration: server.port = 8080
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 224 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl restore --src backups
- productctl backup --dest backups

Fingerprint: 9fe5ba3dbfb3efb31fde5687df26551d

Troubleshooting:
- Symptom: authentication loop
- Error code: E470

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- clear cache directory
- reset config to baseline
- clear cache directory

Fingerprint: eb538b5c84a38879cd738a07a5a4d323

Upgrade Notes:
- From version: 1.3.5
- To version: 1.2.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] verify certificates are valid
[] prepare rollback artifact
[] disable non-essential jobs
[] verify compatibility matrix
[] record current version
[] run pre-upgrade diagnostics
[] notify users
[] stop services cleanly
[] validate free disk space
[] export config snapshot

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restart
- Verification: UI loads and login works

Fingerprint: 142981600df63a88dcfc1f03288cf1a0

Task Overview:
- Task: Restore
- Applies to: Agent
- Prerequisites: admin access

Procedure:
1. Validate environment: ports open
2. Set configuration: feature.flags = true
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 185 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl restart

Fingerprint: a96d8813d1d2369d6eca537433e99d69

Troubleshooting:
- Symptom: authentication loop
- Error code: E593

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- incompatible dependency version
- insufficient permissions

Recovery Actions:
- clear cache directory
- reset config to baseline
- re-run installer with admin rights

Fingerprint: f0c94369537747fa206654be42900e09

Upgrade Notes:
- From version: 1.2.0
- To version: 2.1.2
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] verify certificates are valid
[] notify users
[] confirm maintenance window
[] take a full backup
[] prepare rollback artifact
[] export config snapshot
[] validate free disk space
[] disable non-essential jobs
[] record current version
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl diagnostics
- Verification: no errors in last 5 minutes

Fingerprint: 6f39e70b29681bf841838597cf9c14db

Task Overview:
- Task: Restore
- Applies to: Scheduler
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: feature.flags = true
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 21 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restart
- productctl status

Fingerprint: 77acb3f2bf5971b5a83fe0754a61ecc6

Troubleshooting:
- Symptom: service fails to start
- Error code: E827

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- reset config to baseline
- re-run installer with admin rights
- increase memory limit and redeploy

Fingerprint: 68f92d12a6604cc34d94cd5fb1a6ac67

Upgrade Notes:
- From version: 2.1.2
- To version: 1.4.1
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] stop services cleanly
[] notify users
[] verify compatibility matrix
[] verify certificates are valid
[] confirm maintenance window
[] export config snapshot
[] record current version
[] validate free disk space
[] prepare rollback artifact
[] run pre-upgrade diagnostics

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl restore --src backups
- Verification: UI loads and login works

Fingerprint: 6cd027a9e391b926c2e4c0a708da2bf8

Task Overview:
- Task: Install
- Applies to: Connector
- Prerequisites: admin access

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: server.port = true
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 247 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl restart

Fingerprint: 29971f94cf2a577fcc74e8585df0eba0

Troubleshooting:
- Symptom: authentication loop
- Error code: E679

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- invalid config value
- insufficient permissions

Recovery Actions:
- increase memory limit and redeploy
- increase memory limit and redeploy
- reset config to baseline

Fingerprint: b618eced79558c12594704147f322f14

Upgrade Notes:
- From version: 1.2.0
- To version: 1.3.5
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] export config snapshot
[] disable non-essential jobs
[] take a full backup
[] validate free disk space
[] run pre-upgrade diagnostics
[] notify users
[] verify certificates are valid
[] prepare rollback artifact
[] stop services cleanly
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl diagnostics
- Verification: status endpoint returns OK

Fingerprint: c59ca42a65cd22bad4b0766aeda53229

Task Overview:
- Task: Configure
- Applies to: Control Plane
- Prerequisites: TLS certificates ready

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: cache.size_mb = alpha,beta
3. Restart service: product-agent
4. Verify health: UI loads and login works

Notes:
- Default timeout: 68 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restart
- productctl restore --src backups

Fingerprint: 1146022f66c0d8d483183bb27b75e634

Troubleshooting:
- Symptom: service fails to start
- Error code: E164

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- expired certificate
- invalid config value

Recovery Actions:
- rotate certificate and restart
- increase memory limit and redeploy
- increase memory limit and redeploy

Fingerprint: f0b772cb23f29317e16c588a98e1319f

Upgrade Notes:
- From version: 1.2.0
- To version: 1.3.5
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] take a full backup
[] record current version
[] prepare rollback artifact
[] notify users
[] confirm maintenance window
[] disable non-essential jobs
[] verify certificates are valid
[] run pre-upgrade diagnostics
[] validate free disk space
[] stop services cleanly

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: UI loads and login works

Fingerprint: dde5a9c83c3ac1f2cf43772e205bbad7

Task Overview:
- Task: Backup
- Applies to: Agent
- Prerequisites: admin access

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: telemetry.enabled = tls
3. Restart service: product-ui
4. Verify health: UI loads and login works

Notes:
- Default timeout: 70 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restart
- productctl status

Fingerprint: f3d8cc24dec598c75074167cfe2a260c

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E655

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- incompatible dependency version
- incompatible dependency version

Recovery Actions:
- clear cache directory
- clear cache directory
- clear cache directory

Fingerprint: ee833da3c0ac28b3d498a342b443488f

Upgrade Notes:
- From version: 1.4.1
- To version: 1.4.1
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] validate free disk space
[] verify compatibility matrix
[] notify users
[] prepare rollback artifact
[] export config snapshot
[] stop services cleanly
[] verify certificates are valid

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: b39596ae11fe87d35a66d3042d5ed5e6

Task Overview:
- Task: Upgrade
- Applies to: Scheduler
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: log.level = alpha,beta
3. Restart service: product-ui
4. Verify health: UI loads and login works

Notes:
- Default timeout: 87 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl status
- productctl restore --src backups

Fingerprint: 3b13ad85936dfa076c47a21c80987a63

Troubleshooting:
- Symptom: service fails to start
- Error code: E696

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- expired certificate
- incompatible dependency version

Recovery Actions:
- rotate certificate and restart
- clear cache directory
- clear cache directory

Fingerprint: a7bc0cd96b510be6f4c0e17828552bde

Upgrade Notes:
- From version: 1.2.0
- To version: 1.2.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] verify compatibility matrix
[] export config snapshot
[] verify certificates are valid
[] prepare rollback artifact
[] take a full backup
[] stop services cleanly
[] notify users
[] validate free disk space
[] confirm maintenance window

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl status
- Verification: status endpoint returns OK

Fingerprint: f2f559b71e30a2c22ebc872c3cec4553

Task Overview:
- Task: Configure
- Applies to: Telemetry Module
- Prerequisites: database reachable

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: cache.size_mb = true
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 75 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl restore --src backups
- productctl status

Fingerprint: 52c0db4dafb12aa3521b92c682f57eb5

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E580

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- incompatible dependency version
- expired certificate

Recovery Actions:
- clear cache directory
- rotate certificate and restart
- rotate certificate and restart

Fingerprint: 155f4668b675c17fc9756b178b1d63fd

Upgrade Notes:
- From version: 2.0.0
- To version: 1.3.5
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] export config snapshot
[] verify certificates are valid
[] take a full backup
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] stop services cleanly
[] notify users
[] record current version
[] disable non-essential jobs
[] confirm maintenance window

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: e40bf61da495655de48beab61b43ea62

Task Overview:
- Task: Configure
- Applies to: Control Plane
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: feature.flags = tls
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 134 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl backup --dest backups
- productctl status

Fingerprint: 64cce17112c7f91f48197102c8e4e63f

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E590

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- incompatible dependency version
- expired certificate

Recovery Actions:
- reset config to baseline
- rotate certificate and restart
- re-run installer with admin rights

Fingerprint: 1b01863a7cc38c239996242319e44fb4

Upgrade Notes:
- From version: 2.1.2
- To version: 1.4.1
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] verify compatibility matrix
[] verify certificates are valid
[] export config snapshot
[] stop services cleanly
[] validate free disk space
[] run pre-upgrade diagnostics
[] take a full backup
[] notify users
[] confirm maintenance window

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restore --src backups
- Verification: no errors in last 5 minutes

Fingerprint: 04ae122d79005ad648ec3bd6ecc7912f

Task Overview:
- Task: Rotate Credentials
- Applies to: CLI
- Prerequisites: admin access

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: server.port = true
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 172 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl diagnostics

Fingerprint: 5fceffedf436e0b733f327a849fb7988

Troubleshooting:
- Symptom: service fails to start
- Error code: E928

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- expired certificate
- expired certificate

Recovery Actions:
- increase memory limit and redeploy
- clear cache directory
- clear cache directory

Fingerprint: 16c361bb5d4e2b47d4c88bf4db26a430

Upgrade Notes:
- From version: 2.1.2
- To version: 2.1.2
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] verify certificates are valid
[] stop services cleanly
[] run pre-upgrade diagnostics
[] record current version
[] verify compatibility matrix
[] validate free disk space
[] export config snapshot
[] disable non-essential jobs
[] confirm maintenance window
[] take a full backup

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl backup --dest backups
- Verification: status endpoint returns OK

Fingerprint: 7f0bb67eeea0aa32c90a0929269afc85

Task Overview:
- Task: Install
- Applies to: Web Console
- Prerequisites: admin access

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: auth.mode = info
3. Restart service: product-ui
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 41 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl diagnostics
- productctl restart

Fingerprint: c181416dd15a7a0b51b1cca78ef13ff3

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E215

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- network policy block
- incompatible dependency version

Recovery Actions:
- clear cache directory
- increase memory limit and redeploy
- rotate certificate and restart

Fingerprint: 8299715a07cd4368ce03aa0f3fcf2b92

Upgrade Notes:
- From version: 1.2.0
- To version: 2.0.0
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] export config snapshot
[] disable non-essential jobs
[] record current version
[] take a full backup
[] verify certificates are valid
[] stop services cleanly
[] notify users
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl restore --src backups
- Verification: UI loads and login works

Fingerprint: 7ba655dbf4f5f558eaa030a06f095ca1

Task Overview:
- Task: Tune Performance
- Applies to: Database Layer
- Prerequisites: database reachable

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: server.port = info
3. Restart service: product-agent
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 192 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl status

Fingerprint: 89853b94b2c729c6a4289eef0ff276d0

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E117

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- invalid config value
- incompatible dependency version

Recovery Actions:
- re-run installer with admin rights
- re-run installer with admin rights
- re-run installer with admin rights

Fingerprint: 80888be3bf842316e537e3b605ae696f

Upgrade Notes:
- From version: 1.4.1
- To version: 1.2.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] verify certificates are valid
[] prepare rollback artifact
[] verify compatibility matrix
[] run pre-upgrade diagnostics
[] take a full backup
[] validate free disk space
[] stop services cleanly
[] record current version
[] notify users

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl restore --src backups
- Verification: metrics endpoint responds

Fingerprint: f717be86d14eb17202cc1f0bf23aecfe

Task Overview:
- Task: Upgrade
- Applies to: Agent
- Prerequisites: admin access

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: server.port = 40
3. Restart service: telemetry
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 103 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl diagnostics
- productctl diagnostics

Fingerprint: 561c8ffa3d6f36eda2e11b766f25b90d

Troubleshooting:
- Symptom: authentication loop
- Error code: E535

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- network policy block
- invalid config value

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- increase memory limit and redeploy

Fingerprint: 024b4bed2b9dc92e4e16e4a60969ebf5

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] stop services cleanly
[] take a full backup
[] verify compatibility matrix
[] verify certificates are valid
[] notify users
[] record current version
[] disable non-essential jobs
[] validate free disk space

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: 545e73baa4b75d8d5fd09601382eb3ce

Task Overview:
- Task: Rotate Credentials
- Applies to: CLI
- Prerequisites: supported OS image

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: db.pool_max = 512
3. Restart service: telemetry
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 175 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl restart
- productctl restart

Fingerprint: 7f930ef465eacd0e10ca96ec84cf76bf

Troubleshooting:
- Symptom: authentication loop
- Error code: E761

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- incompatible dependency version
- invalid config value

Recovery Actions:
- rotate certificate and restart
- rotate certificate and restart
- rotate certificate and restart

Fingerprint: 71af22f5c3d6c7cba7a06feea42b15e4

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] notify users
[] take a full backup
[] record current version
[] verify compatibility matrix
[] validate free disk space
[] verify certificates are valid
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] prepare rollback artifact

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl status
- Verification: status endpoint returns OK

Fingerprint: 0a34e807a590b5dc8b7e22fce2199dc6

Task Overview:
- Task: Configure
- Applies to: Control Plane
- Prerequisites: database reachable

Procedure:
1. Validate environment: ports open
2. Set configuration: auth.mode = 512
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 74 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl status
- productctl restart

Fingerprint: 66111abfd988653a787493f049b0b676

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E449

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- insufficient permissions
- network policy block

Recovery Actions:
- re-run installer with admin rights
- rotate certificate and restart
- re-run installer with admin rights

Fingerprint: 196783c87330d3f63b5313b9989f9f2a

Upgrade Notes:
- From version: 1.4.1
- To version: 2.1.2
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] verify certificates are valid
[] confirm maintenance window
[] take a full backup
[] run pre-upgrade diagnostics
[] validate free disk space
[] notify users
[] export config snapshot
[] record current version
[] prepare rollback artifact
[] stop services cleanly

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl diagnostics
- Verification: metrics endpoint responds

Fingerprint: 11916bccc5de4931547eef0980cf7b48

Task Overview:
- Task: Install
- Applies to: CLI
- Prerequisites: supported OS image

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: log.level = 8080
3. Restart service: product-ui
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 91 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl status
- productctl diagnostics

Fingerprint: a42b5228d84fcddd3de61d4f3b443cdf

Troubleshooting:
- Symptom: authentication loop
- Error code: E664

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- incompatible dependency version
- incompatible dependency version

Recovery Actions:
- reset config to baseline
- clear cache directory
- rotate certificate and restart

Fingerprint: 9db237f3c9885c115eb852ea8bdde3b5

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] notify users
[] take a full backup
[] validate free disk space
[] prepare rollback artifact
[] verify compatibility matrix
[] record current version
[] verify certificates are valid
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] stop services cleanly

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl status
- Verification: UI loads and login works

Fingerprint: 5b47a6e21391a05d904a33b322270b96

Task Overview:
- Task: Configure
- Applies to: Control Plane
- Prerequisites: DNS configured

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: log.level = 8080
3. Restart service: product-api
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 66 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl restore --src backups
- productctl backup --dest backups

Fingerprint: 795144e60a3769d54ca7c0820b8d9791

Troubleshooting:
- Symptom: authentication loop
- Error code: E678

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- incompatible dependency version
- insufficient permissions

Recovery Actions:
- clear cache directory
- clear cache directory
- reset config to baseline

Fingerprint: 3c330f688f187ad2fd9550b6967a2026

Upgrade Notes:
- From version: 2.0.0
- To version: 1.2.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] notify users
[] export config snapshot
[] verify certificates are valid
[] take a full backup
[] validate free disk space
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: dd9b4b4e208983983c4caaf2dea9d815

Task Overview:
- Task: Backup
- Applies to: Control Plane
- Prerequisites: DNS configured

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: db.pool_max = 8080
3. Restart service: product-agent
4. Verify health: UI loads and login works

Notes:
- Default timeout: 106 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl backup --dest backups

Fingerprint: e2cef1f9977cd1b0b733e7a04bdeec42

Troubleshooting:
- Symptom: authentication loop
- Error code: E864

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- re-run installer with admin rights
- increase memory limit and redeploy
- re-run installer with admin rights

Fingerprint: 36618210d7409c60a9c4ba353d33e9fa

Upgrade Notes:
- From version: 1.2.0
- To version: 1.3.5
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] verify compatibility matrix
[] disable non-essential jobs
[] export config snapshot
[] notify users
[] validate free disk space
[] confirm maintenance window
[] take a full backup
[] stop services cleanly
[] verify certificates are valid

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl backup --dest backups
- Verification: status endpoint returns OK

Fingerprint: 7d508990cb56e4c3247f225961c0de3f

Task Overview:
- Task: Backup
- Applies to: Control Plane
- Prerequisites: admin access

Procedure:
1. Validate environment: ports open
2. Set configuration: server.port = 8080
3. Restart service: telemetry
4. Verify health: UI loads and login works

Notes:
- Default timeout: 163 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl restore --src backups
- productctl diagnostics

Fingerprint: 1445190ba9c69a29ebf37604a4b40e3e

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E126

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- expired certificate
- expired certificate

Recovery Actions:
- rotate certificate and restart
- re-run installer with admin rights
- re-run installer with admin rights

Fingerprint: a5c6645c5e15ce9e14787a6d92cb1d44

Upgrade Notes:
- From version: 1.4.1
- To version: 1.2.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] confirm maintenance window
[] stop services cleanly
[] verify compatibility matrix
[] record current version
[] validate free disk space
[] disable non-essential jobs
[] take a full backup
[] export config snapshot

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl restart
- Verification: metrics endpoint responds

Fingerprint: cf03877f7ea8584e0409ea201c5ab5a2

Task Overview:
- Task: Restore
- Applies to: Scheduler
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: log.level = true
3. Restart service: telemetry
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 63 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl status
- productctl restore --src backups

Fingerprint: 65f9fffeeade44a05c54fa7ef23893cf

Troubleshooting:
- Symptom: high CPU usage
- Error code: E709

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- network policy block
- network policy block

Recovery Actions:
- reset config to baseline
- rotate certificate and restart
- clear cache directory

Fingerprint: ed1df5a515e80aed95b9a108a65a266c

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] confirm maintenance window
[] notify users
[] validate free disk space
[] verify compatibility matrix
[] record current version
[] export config snapshot
[] take a full backup

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl status
- Verification: metrics endpoint responds

Fingerprint: 77d9b500906d2c9058b66a9cefe61c31

Task Overview:
- Task: Rotate Credentials
- Applies to: Database Layer
- Prerequisites: supported OS image

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: telemetry.enabled = 40
3. Restart service: product-api
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 146 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restart
- productctl restart

Fingerprint: 45b2fb0c98429519acd9d2c4ab674c7b

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E213

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- network policy block
- expired certificate

Recovery Actions:
- rotate certificate and restart
- clear cache directory
- re-run installer with admin rights

Fingerprint: 29d8e95a5b4ac28937067c6e0fe4dbb9

Upgrade Notes:
- From version: 1.3.5
- To version: 2.0.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] take a full backup
[] prepare rollback artifact
[] disable non-essential jobs
[] validate free disk space
[] record current version
[] stop services cleanly
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] export config snapshot
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl diagnostics
- Verification: UI loads and login works

Fingerprint: b3f23407ac86d5574da98a4bdc3d9a8b

Task Overview:
- Task: Upgrade
- Applies to: Control Plane
- Prerequisites: supported OS image

Procedure:
1. Validate environment: disk >= 10GB free
2. Set configuration: telemetry.enabled = 8080
3. Restart service: product-ui
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 165 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl backup --dest backups
- productctl diagnostics

Fingerprint: f7f3421ad0917c561fba1d2bcbf561cc

Troubleshooting:
- Symptom: service fails to start
- Error code: E595

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- insufficient permissions
- expired certificate

Recovery Actions:
- increase memory limit and redeploy
- rotate certificate and restart
- increase memory limit and redeploy

Fingerprint: 384e96685c12a677d82933565caea42f

Upgrade Notes:
- From version: 2.0.0
- To version: 1.4.1
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] export config snapshot
[] run pre-upgrade diagnostics
[] record current version
[] confirm maintenance window
[] stop services cleanly
[] verify certificates are valid
[] disable non-essential jobs
[] verify compatibility matrix
[] notify users
[] prepare rollback artifact

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: 2e33fd9779328e85a9f04bbfdb9ed4db

Task Overview:
- Task: Restore
- Applies to: Agent
- Prerequisites: supported OS image

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: server.port = 512
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 108 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl status

Fingerprint: cf6d99f655a80281d1061801f6368dca

Troubleshooting:
- Symptom: service fails to start
- Error code: E238

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- incompatible dependency version
- network policy block

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- rotate certificate and restart

Fingerprint: ed016ec4e25ac84063981768935cbc6a

Upgrade Notes:
- From version: 2.0.0
- To version: 1.4.1
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] confirm maintenance window
[] take a full backup
[] prepare rollback artifact
[] stop services cleanly
[] verify certificates are valid
[] validate free disk space
[] export config snapshot
[] notify users
[] record current version

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: 9469ea7d327d89234880b090a2e6a1b1

Task Overview:
- Task: Rotate Credentials
- Applies to: Agent
- Prerequisites: admin access

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: telemetry.enabled = 8080
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 93 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl diagnostics
- productctl diagnostics

Fingerprint: 3f869f774b19aa12598dce11ab85ef37

Troubleshooting:
- Symptom: authentication loop
- Error code: E680

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- invalid config value
- expired certificate

Recovery Actions:
- re-run installer with admin rights
- rotate certificate and restart
- clear cache directory

Fingerprint: c101c1c31a4ee6cf7cf376b9c31b3a3e

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] take a full backup
[] stop services cleanly
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] disable non-essential jobs
[] validate free disk space
[] export config snapshot
[] verify certificates are valid
[] confirm maintenance window
[] notify users

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restart
- Verification: metrics endpoint responds

Fingerprint: 575b8da1ed9a2eabe350378cfc2e59a0

Task Overview:
- Task: Tune Performance
- Applies to: CLI
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: feature.flags = 512
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 108 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl diagnostics
- productctl restore --src backups

Fingerprint: bd87b9951157f4e1e33b8f0c43fdb5ef

Troubleshooting:
- Symptom: UI shows 500 error
- Error code: E595

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- expired certificate
- insufficient permissions

Recovery Actions:
- clear cache directory
- rotate certificate and restart
- reset config to baseline

Fingerprint: a79346873865ca4c7bfcb69b8d5e2b65

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] take a full backup
[] stop services cleanly
[] record current version
[] validate free disk space
[] disable non-essential jobs
[] prepare rollback artifact
[] export config snapshot
[] notify users
[] verify compatibility matrix
[] confirm maintenance window

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl diagnostics
- Verification: status endpoint returns OK

Fingerprint: 1430bb5589e8f3583752b6d140d726e3

Task Overview:
- Task: Install
- Applies to: CLI
- Prerequisites: admin access

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: log.level = 512
3. Restart service: telemetry
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 160 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl backup --dest backups

Fingerprint: 38f91f66fe872c5fe3637e7aa432cebc

Troubleshooting:
- Symptom: service fails to start
- Error code: E810

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- expired certificate
- invalid config value

Recovery Actions:
- rotate certificate and restart
- rotate certificate and restart
- clear cache directory

Fingerprint: 93c2844ac624796e3863563d5e79b504

Upgrade Notes:
- From version: 1.2.0
- To version: 1.4.1
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] validate free disk space
[] confirm maintenance window
[] run pre-upgrade diagnostics
[] take a full backup
[] record current version
[] export config snapshot
[] notify users
[] disable non-essential jobs
[] verify certificates are valid
[] stop services cleanly

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restart
- Verification: metrics endpoint responds

Fingerprint: 76930c4a4ca6093a6701e4bea9df1925

Task Overview:
- Task: Collect Diagnostics
- Applies to: Connector
- Prerequisites: DNS configured

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: db.pool_max = 40
3. Restart service: product-agent
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 62 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl backup --dest backups

Fingerprint: c717f074e175342852d5cd763714d024

Troubleshooting:
- Symptom: service fails to start
- Error code: E591

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- insufficient permissions
- expired certificate

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- clear cache directory

Fingerprint: 5129a8c472b68200bf1775030c621a40

Upgrade Notes:
- From version: 2.0.0
- To version: 2.1.2
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] confirm maintenance window
[] stop services cleanly
[] validate free disk space
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] export config snapshot
[] take a full backup
[] record current version
[] notify users
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: status endpoint returns OK

Fingerprint: 334e0e4b98758b830ed6a68a483f17cc

Task Overview:
- Task: Rotate Credentials
- Applies to: Web Console
- Prerequisites: TLS certificates ready

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: log.level = tls
3. Restart service: product-agent
4. Verify health: UI loads and login works

Notes:
- Default timeout: 52 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl status
- productctl backup --dest backups

Fingerprint: 1ed28261285f357af83217245f5b7600

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E451

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- incompatible dependency version
- invalid config value

Recovery Actions:
- increase memory limit and redeploy
- rotate certificate and restart
- rotate certificate and restart

Fingerprint: 9e0aa760f92f5a45a6c65156cb514a98

Upgrade Notes:
- From version: 1.3.5
- To version: 1.2.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] prepare rollback artifact
[] confirm maintenance window
[] verify compatibility matrix
[] record current version
[] disable non-essential jobs
[] export config snapshot
[] take a full backup
[] verify certificates are valid
[] notify users

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl diagnostics
- Verification: no errors in last 5 minutes

Fingerprint: 7566b5cb7e8ed5babe5bcfba56a6b541

Task Overview:
- Task: Backup
- Applies to: Database Layer
- Prerequisites: DNS configured

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: server.port = info
3. Restart service: telemetry
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 47 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl restart
- productctl diagnostics

Fingerprint: 4562a89fb1616dea7fa5a88bedf17912

Troubleshooting:
- Symptom: high CPU usage
- Error code: E530

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- network policy block
- invalid config value

Recovery Actions:
- re-run installer with admin rights
- increase memory limit and redeploy
- increase memory limit and redeploy

Fingerprint: 46c839eeb11d970eb141406bd0d3aa72

Upgrade Notes:
- From version: 1.2.0
- To version: 1.4.1
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] export config snapshot
[] confirm maintenance window
[] prepare rollback artifact
[] disable non-essential jobs
[] stop services cleanly
[] notify users
[] record current version
[] validate free disk space
[] verify certificates are valid

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl status
- Verification: metrics endpoint responds

Fingerprint: 882ec9cd2c617087c2c95a58c9a51c7a

Task Overview:
- Task: Upgrade
- Applies to: Scheduler
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: log.level = info
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 99 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl diagnostics

Fingerprint: 076011bcf5082f441cd5809d1b6cac78

Troubleshooting:
- Symptom: high CPU usage
- Error code: E436

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- invalid config value
- invalid config value

Recovery Actions:
- rotate certificate and restart
- re-run installer with admin rights
- re-run installer with admin rights

Fingerprint: a32d0e818398736b51d508975e16fabd

Upgrade Notes:
- From version: 2.1.2
- To version: 1.2.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] record current version
[] confirm maintenance window
[] take a full backup
[] verify compatibility matrix
[] run pre-upgrade diagnostics
[] export config snapshot
[] verify certificates are valid
[] prepare rollback artifact
[] disable non-essential jobs
[] stop services cleanly

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl backup --dest backups
- Verification: no errors in last 5 minutes

Fingerprint: f94469721ccae83f0da7d528e40d8eeb

Task Overview:
- Task: Configure
- Applies to: Telemetry Module
- Prerequisites: DNS configured

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: auth.mode = true
3. Restart service: product-api
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 83 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl restore --src backups

Fingerprint: 90bf1f4dbeb0b3c57c61b7e2493576cb

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E662

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- insufficient permissions
- invalid config value

Recovery Actions:
- reset config to baseline
- clear cache directory
- re-run installer with admin rights

Fingerprint: e35b5463c844196ce3ff76834cdcabd9

Upgrade Notes:
- From version: 2.1.2
- To version: 1.3.5
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] stop services cleanly
[] take a full backup
[] validate free disk space
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] record current version
[] disable non-essential jobs
[] export config snapshot
[] verify compatibility matrix
[] notify users

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl backup --dest backups
- Verification: no errors in last 5 minutes

Fingerprint: 704d2b3a3b7abe26358dad8d00eb9a6a

Task Overview:
- Task: Install
- Applies to: CLI
- Prerequisites: supported OS image

Procedure:
1. Validate environment: ports open
2. Set configuration: db.pool_max = 8080
3. Restart service: product-agent
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 148 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl status
- productctl backup --dest backups

Fingerprint: 1acdaa78658a2360ad8541f310e9c023

Troubleshooting:
- Symptom: high CPU usage
- Error code: E997

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- invalid config value
- insufficient permissions

Recovery Actions:
- reset config to baseline
- clear cache directory
- increase memory limit and redeploy

Fingerprint: 4ce2f824124a725cc0b270728cd4bd40

Upgrade Notes:
- From version: 1.3.5
- To version: 1.4.1
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] export config snapshot
[] verify certificates are valid
[] validate free disk space
[] verify compatibility matrix
[] confirm maintenance window
[] notify users
[] stop services cleanly
[] disable non-essential jobs
[] take a full backup
[] run pre-upgrade diagnostics

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restart
- Verification: no errors in last 5 minutes

Fingerprint: e09d49af788b3a76dd5d380057577dc5

Task Overview:
- Task: Backup
- Applies to: Telemetry Module
- Prerequisites: supported OS image

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: db.pool_max = tls
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 104 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl status
- productctl restart

Fingerprint: d6855334fff3adcf5d18790b5bc01eed

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E289

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- expired certificate
- invalid config value

Recovery Actions:
- reset config to baseline
- increase memory limit and redeploy
- clear cache directory

Fingerprint: ee5d7b425424763445cce6613e984e34

Upgrade Notes:
- From version: 2.1.2
- To version: 1.2.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] confirm maintenance window
[] validate free disk space
[] stop services cleanly
[] verify compatibility matrix
[] export config snapshot
[] verify certificates are valid
[] disable non-essential jobs
[] run pre-upgrade diagnostics
[] notify users
[] record current version

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl restore --src backups
- Verification: UI loads and login works

Fingerprint: e4709d0a61af32c8970773568a5c48fb

Task Overview:
- Task: Restore
- Applies to: Web Console
- Prerequisites: TLS certificates ready

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: log.level = alpha,beta
3. Restart service: product-ui
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 143 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl status
- productctl backup --dest backups

Fingerprint: cef59e37f071005bdae6687181d74a64

Troubleshooting:
- Symptom: authentication loop
- Error code: E104

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- increase memory limit and redeploy
- clear cache directory
- increase memory limit and redeploy

Fingerprint: 9e4130092fcd4416532616ce6a81262e

Upgrade Notes:
- From version: 1.4.1
- To version: 1.2.0
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] verify certificates are valid
[] take a full backup
[] confirm maintenance window
[] stop services cleanly
[] notify users
[] record current version
[] export config snapshot
[] validate free disk space
[] prepare rollback artifact

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: 8c905fb122daf0ae4bcd8fccaa4ff3d0

Task Overview:
- Task: Upgrade
- Applies to: CLI
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: telemetry.enabled = 512
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 190 seconds.
- Log bundle path: /etc/product

Command Snippets:
- productctl backup --dest backups
- productctl backup --dest backups

Fingerprint: c0fe53571b7084ad230a4ca2b487aed8

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E538

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- network policy block
- expired certificate

Recovery Actions:
- re-run installer with admin rights
- reset config to baseline
- increase memory limit and redeploy

Fingerprint: c4d76526b9fba590e9704a511967ba68

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] take a full backup
[] disable non-essential jobs
[] run pre-upgrade diagnostics
[] verify certificates are valid
[] export config snapshot
[] confirm maintenance window
[] verify compatibility matrix
[] notify users
[] validate free disk space
[] record current version

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl backup --dest backups
- Verification: UI loads and login works

Fingerprint: 49f018067e40e403c691c203a5bc3c39

Task Overview:
- Task: Backup
- Applies to: Scheduler
- Prerequisites: admin access

Procedure:
1. Validate environment: ports open
2. Set configuration: log.level = 8080
3. Restart service: product-api
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 90 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl diagnostics
- productctl diagnostics

Fingerprint: 12734e2be1eadc58f37444ff23a31ca3

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E607

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- incompatible dependency version
- incompatible dependency version

Recovery Actions:
- reset config to baseline
- re-run installer with admin rights
- re-run installer with admin rights

Fingerprint: e4e08c7e0d70a873a81e31125758e9fd

Upgrade Notes:
- From version: 1.4.1
- To version: 1.3.5
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] run pre-upgrade diagnostics
[] validate free disk space
[] stop services cleanly
[] verify compatibility matrix
[] export config snapshot
[] disable non-essential jobs
[] verify certificates are valid
[] take a full backup
[] prepare rollback artifact
[] record current version

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl diagnostics
- Verification: metrics endpoint responds

Fingerprint: 7e450f26b850c9261c3f3169da10533b

Task Overview:
- Task: Install
- Applies to: Web Console
- Prerequisites: supported OS image

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: feature.flags = info
3. Restart service: product-agent
4. Verify health: UI loads and login works

Notes:
- Default timeout: 178 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl diagnostics
- productctl status

Fingerprint: c18314a7c4c49c664d0668bb1ec4d85c

Troubleshooting:
- Symptom: authentication loop
- Error code: E656

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- network policy block
- expired certificate

Recovery Actions:
- reset config to baseline
- re-run installer with admin rights
- re-run installer with admin rights

Fingerprint: 52ed79394a1f9c94d9b2c2d7a7e57875

Upgrade Notes:
- From version: 1.3.5
- To version: 2.1.2
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] record current version
[] export config snapshot
[] stop services cleanly
[] run pre-upgrade diagnostics
[] take a full backup
[] verify compatibility matrix
[] validate free disk space
[] notify users
[] disable non-essential jobs
[] confirm maintenance window

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: metrics endpoint responds

Fingerprint: 8b5ac09eb5f7db9970523afa4926ddda

Task Overview:
- Task: Install
- Applies to: Scheduler
- Prerequisites: database reachable

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: auth.mode = 8080
3. Restart service: product-agent
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 219 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl restart

Fingerprint: b059968edc58e876f9cb7819a25fa9b1

Troubleshooting:
- Symptom: authentication loop
- Error code: E656

Decision Tree:
1. If disk is low, free space under C:\Product\config.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- invalid config value
- invalid config value
- insufficient permissions

Recovery Actions:
- re-run installer with admin rights
- clear cache directory
- reset config to baseline

Fingerprint: efd92d41e132159c9cf30437c669c44c

Upgrade Notes:
- From version: 1.4.1
- To version: 1.4.1
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] take a full backup
[] prepare rollback artifact
[] verify compatibility matrix
[] stop services cleanly
[] verify certificates are valid
[] record current version
[] export config snapshot
[] run pre-upgrade diagnostics
[] notify users
[] confirm maintenance window

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: UI loads and login works

Fingerprint: b7920ec6aabf7518bf8d41b0bfb44fcd

Task Overview:
- Task: Configure
- Applies to: Database Layer
- Prerequisites: admin access

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: server.port = 40
3. Restart service: telemetry
4. Verify health: no errors in last 5 minutes

Notes:
- Default timeout: 181 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl restart

Fingerprint: 0580fc3e17f356562d5ea784ce3abe9b

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E565

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- clear cache directory
- clear cache directory
- rotate certificate and restart

Fingerprint: 62a6cd847678cd3737d383192d4bc3bf

Upgrade Notes:
- From version: 2.0.0
- To version: 2.1.2
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] validate free disk space
[] disable non-essential jobs
[] export config snapshot
[] verify compatibility matrix
[] confirm maintenance window
[] notify users
[] stop services cleanly
[] prepare rollback artifact
[] verify certificates are valid
[] take a full backup

Rollback Plan:
- Backup artifact: upgrade log bundle
- Restore command: productctl status
- Verification: UI loads and login works

Fingerprint: d8dd69aa438dc9c76bb3bf964b339b07

Task Overview:
- Task: Tune Performance
- Applies to: Connector
- Prerequisites: database reachable

Procedure:
1. Validate environment: clock sync enabled
2. Set configuration: cache.size_mb = 40
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 48 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl restore --src backups
- productctl restart

Fingerprint: 79abe1c20700e978041a715b58564cf8

Troubleshooting:
- Symptom: high CPU usage
- Error code: E670

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- expired certificate
- incompatible dependency version

Recovery Actions:
- clear cache directory
- reset config to baseline
- rotate certificate and restart

Fingerprint: 07a08f376544a55ef104b3b0fbd7ad62

Upgrade Notes:
- From version: 1.4.1
- To version: 1.3.5
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] verify certificates are valid
[] verify compatibility matrix
[] confirm maintenance window
[] take a full backup
[] export config snapshot
[] disable non-essential jobs
[] run pre-upgrade diagnostics
[] notify users
[] validate free disk space
[] stop services cleanly

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl diagnostics
- Verification: metrics endpoint responds

Fingerprint: fb7e677333e9f8c70d715d530907c3d5

Task Overview:
- Task: Backup
- Applies to: Web Console
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: telemetry.enabled = tls
3. Restart service: product-agent
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 220 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl status
- productctl diagnostics

Fingerprint: 01af10f20f6d31f3cc6c9ed8adaff209

Troubleshooting:
- Symptom: high CPU usage
- Error code: E687

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- expired certificate
- incompatible dependency version

Recovery Actions:
- re-run installer with admin rights
- rotate certificate and restart
- re-run installer with admin rights

Fingerprint: 50953304594ef3fb58126260a6ae5659

Upgrade Notes:
- From version: 2.1.2
- To version: 1.3.5
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] disable non-essential jobs
[] notify users
[] stop services cleanly
[] take a full backup
[] run pre-upgrade diagnostics
[] confirm maintenance window
[] export config snapshot
[] record current version
[] validate free disk space

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl diagnostics
- Verification: UI loads and login works

Fingerprint: 41b8b75810dba5b9cc0462c0271d46f4

Task Overview:
- Task: Upgrade
- Applies to: Agent
- Prerequisites: supported OS image

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: server.port = 40
3. Restart service: product-agent
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 68 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl restore --src backups
- productctl backup --dest backups

Fingerprint: 6527f692053972b6d5f7485824aefd3c

Troubleshooting:
- Symptom: authentication loop
- Error code: E137

Decision Tree:
1. If disk is low, free space under /var/log/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- network policy block
- network policy block
- invalid config value

Recovery Actions:
- reset config to baseline
- clear cache directory
- re-run installer with admin rights

Fingerprint: 6d3a0563c39a2482ad56a436e8866bfd

Upgrade Notes:
- From version: 1.3.5
- To version: 1.2.0
- Compatibility: requires .NET runtime

Pre-Upgrade Checklist:
[] verify certificates are valid
[] run pre-upgrade diagnostics
[] export config snapshot
[] take a full backup
[] validate free disk space
[] verify compatibility matrix
[] prepare rollback artifact
[] notify users
[] record current version
[] disable non-essential jobs

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl backup --dest backups
- Verification: no errors in last 5 minutes

Fingerprint: 0a2f5d2f37becccadf8dc64cfa311890

Task Overview:
- Task: Backup
- Applies to: Web Console
- Prerequisites: DNS configured

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: telemetry.enabled = 8080
3. Restart service: product-api
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 213 seconds.
- Log bundle path: C:\Product\config

Command Snippets:
- productctl restore --src backups
- productctl status

Fingerprint: d45aa94b80a007118c9ae23deb69edbe

Troubleshooting:
- Symptom: missing data in dashboard
- Error code: E236

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- insufficient permissions
- insufficient permissions
- incompatible dependency version

Recovery Actions:
- rotate certificate and restart
- re-run installer with admin rights
- rotate certificate and restart

Fingerprint: ee59d616d8005178cabaa641a597fd87

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: supports Windows Server 2022

Pre-Upgrade Checklist:
[] disable non-essential jobs
[] confirm maintenance window
[] record current version
[] prepare rollback artifact
[] stop services cleanly
[] verify compatibility matrix
[] export config snapshot
[] run pre-upgrade diagnostics
[] notify users
[] validate free disk space

Rollback Plan:
- Backup artifact: backup manifest
- Restore command: productctl restore --src backups
- Verification: status endpoint returns OK

Fingerprint: aed3d2d25dcf806b676b9ffc7b277ea2

Task Overview:
- Task: Collect Diagnostics
- Applies to: Control Plane
- Prerequisites: admin access

Procedure:
1. Validate environment: outbound HTTPS allowed
2. Set configuration: server.port = alpha,beta
3. Restart service: telemetry
4. Verify health: metrics endpoint responds

Notes:
- Default timeout: 112 seconds.
- Log bundle path: /var/log/product

Command Snippets:
- productctl diagnostics
- productctl diagnostics

Fingerprint: 74fd5fe082b856fbc3fbcaa694cee32b

Troubleshooting:
- Symptom: authentication loop
- Error code: E221

Decision Tree:
1. If disk is low, free space under /etc/product.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- expired certificate
- invalid config value
- insufficient permissions

Recovery Actions:
- clear cache directory
- increase memory limit and redeploy
- reset config to baseline

Fingerprint: 49e7ff284d4302607e6410636c637679

Upgrade Notes:
- From version: 2.1.2
- To version: 2.0.0
- Compatibility: requires PostgreSQL 14+

Pre-Upgrade Checklist:
[] record current version
[] stop services cleanly
[] confirm maintenance window
[] take a full backup
[] prepare rollback artifact
[] run pre-upgrade diagnostics
[] validate free disk space
[] verify certificates are valid
[] notify users
[] verify compatibility matrix

Rollback Plan:
- Backup artifact: config snapshot file
- Restore command: productctl restore --src backups
- Verification: no errors in last 5 minutes

Fingerprint: 729ea084989304be4e4fe0ad36b1ffe9

Task Overview:
- Task: Collect Diagnostics
- Applies to: Telemetry Module
- Prerequisites: supported OS image

Procedure:
1. Validate environment: RAM >= 8GB
2. Set configuration: cache.size_mb = tls
3. Restart service: product-ui
4. Verify health: status endpoint returns OK

Notes:
- Default timeout: 104 seconds.
- Log bundle path: C:\ProgramData\Product\logs

Command Snippets:
- productctl diagnostics
- productctl status

Fingerprint: 6d73f089fc3e5439b0f2263a7c06595d

Troubleshooting:
- Symptom: service fails to start
- Error code: E787

Decision Tree:
1. If disk is low, free space under C:\ProgramData\Product\logs.
2. If config mismatch, compare snapshot with baseline profile.
3. If network issue, verify DNS and firewall rules.
4. If authentication failure, rotate credentials and re-test.

Likely Causes:
- incompatible dependency version
- expired certificate
- insufficient permissions

Recovery Actions:
- increase memory limit and redeploy
- clear cache directory
- rotate certificate and restart

Fingerprint: fce4ff403cdeb488ff349a8e75a72eed

Upgrade Notes:
- From version: 1.4.1
- To version: 2.0.0
- Compatibility: supports Ubuntu 22.04

Pre-Upgrade Checklist:
[] verify compatibility matrix
[] validate free disk space
[] stop services cleanly
[] verify certificates are valid
[] run pre-upgrade diagnostics
[] notify users
[] prepare rollback artifact
[] confirm maintenance window
[] take a full backup
[] disable non-essential jobs

Rollback Plan:
- Backup artifact: diagnostic archive
- Restore command: productctl backup --dest backups
- Verification: UI loads and login works

Fingerprint: b5435c745f32e609139f1776a1c20d38